Merchant Payouts API - Non-Functional Requirements (Draft 2)

1. Context
The Payouts API lets merchants request payouts of their settled balance to a UK bank account. It is called by merchant back-office systems and by our own dashboard. Peak volume is Black Friday, when traffic is roughly five times a normal day.

2. Performance
NFR-P1 The API must respond in under 300 ms for 95% of requests.
NFR-P2 The nightly payout batch must complete before 06:00 so funds arrive the same day.
NFR-P3 The platform must handle 2 million payout requests per day.
NFR-P4 The dashboard should load quickly.

3. Availability and recovery

NFR-A1 The API must be available 99.95% of the time per month, excluding planned maintenance.

NFR-A2 In a disaster, the service must be restored within 1 hour with no more than 15 minutes of data loss.

NFR-A3 Webhooks to merchants are retried if delivery fails.

4. Security

NFR-S1 Merchants authenticate with API keys.

NFR-S2 Each merchant is limited to 100 requests per second.

NFR-S3 Full card numbers are never stored; bank account numbers are encrypted.

NFR-S4 The API must comply with PCI DSS.

5. Data and audit

NFR-D1 Every payout request and status change is recorded in an audit log kept for 7 years.

NFR-D2 Card and bank details must be masked in application logs.

NFR-D3 Merchants can export their payout history as CSV.

NFR-D4 A payout, once confirmed, is never processed twice.